

Third Terminal Examination - 2079

Class : XI

Subject: Chemistry (3011)

Full Marks: 75

Time: 3 hrs.

SET 'B'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

- The number of molecules in 36 mg of water is
 - 12×10^{23}
 - 6.023×10^{23}
 - 12.04×10^{20}
 - 12×10^{21}
- Which of the following sets of quantum number is possible?

n	l	m	s
a. 3	2	3	$\frac{1}{2}$
b. 3	1	0	$+\frac{1}{2}$
c. 3	3	1	$-\frac{1}{2}$
d. 2	1	-2	$\frac{1}{2}$
- When combined, to form water the volumes of hydrogen and oxygen gas bear a simple ratio of 2:1. This is explained by.....
 - Law of Multiple proportions
 - Avogadro's Law
 - Law of Definite Proportion
 - Gay Lussac's Law of Gaseous volume

4. The ionization energy of nitrogen is more than that of oxygen because
 - a. of the extra stability of half-filled p-orbitals in nitrogen
 - b. of the smaller size of nitrogen
 - c. the former contains less number of electrons
 - d. the former is less electronegative
5. For the manufacture of ammonia by the reaction,

$$\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3 + 22.8 \text{ kcal}$$
 the favorable conditions are
 - a. Low temperature, low pressure and catalyst
 - b. Low temperature, high pressure and catalyst
 - c. High temperature, low pressure, catalyst
 - d. High temperature, high pressure, catalyst
6. The adsorption of hydrogen by Pd is
 - a. reduction
 - b. occlusion
 - c. hydrogenation
 - d. dehydrogenation
7. During the manufacture of sulphuric acid by contact process, arsenic oxide is absorbed by
 - a. V_2O_5
 - b. $\text{Fe}(\text{OH})_3$
 - c. Water vapour
 - d. conc. H_2SO_4
8. In the froth floatation process for the purification of ores, the ore particles float because
 - a. They are light
 - b. Their surface is hydrophobic., i.e, not easily wetted by water
 - c. They bear electrostatic charge
 - d. They are insoluble
9. Which of the following is the functional group of acid amide?
 - a. $-\text{COCl}$
 - b. $-\text{COOR}$
 - c. $-(\text{CO})_2\text{O}$
 - d. $-\text{CONH}_2$
10. Cetane rating is used to indicate the quality of as a fuel.
 - a. kerosene
 - b. petrol
 - c. diesel
 - d. LPG

11.
$$\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3 - \text{CH}_2 - \text{C} - \text{CH}_3 \\ | \\ \text{CH}_3 \end{array} \rightarrow \text{IUPAC name is}$$
 - a. 2, 2 – dimethyl butane
 - b. Neo – hexane
 - c. 2, 2 dipropyl propane
 - d. hexane

Group - B

Give short answer to the following questions.

[8 × 5 = 40]

1. a. Iron is a very important micronutrient required for the proper functioning of the human body. It is the main constituent of hemoglobin that occurs as Fe^{++} which helps for transferring oxygen from the lungs to different tissues. [2+1]
i. State Hund's rule and calculate the number of unpaired electrons in Fe^{++} ion.
ii. Determine the value of all quantum numbers of valence electrons of a copper atom?
b. Write the differences between orbit and orbitals. [2]
2. a. Balance the given redox reaction with the help of the oxidation number method or ion-electron method. [3+2]
$$\text{NaOH} + \text{Br}_2 \longrightarrow \text{NaBr} + \text{NaBrO}_3 + \text{H}_2\text{O}$$

b. Calculate the oxidation number of the underline element.
i) $\text{K}_4[\underline{\text{Fe}}(\text{CN})_6]$ ii) $\text{K}_2\underline{\text{Cr}}_2\text{O}_7$
3. Amedeo Avogadro, an Italian scientist in 1831 modified Berzilius hypothesis by introducing the term molecule. According to him a molecule is the smallest particle of an element or a compound which have an independent existence. This is the basis of Avogadros's hypothesis.
a. State the hypothesis. [1]
b. Show that, molecular weight = $2 \times$ Vapour density [2]
c. Calculate the mass of 120 cc of nitrogen at NTP. How many number of molecules are present in it? [2]

OR

- a. A number of anomalies were found in Mendeleev's periodic table. Henry Moseley addresses these anomalies and gave his modern periodic table in 1913 which successfully explains the periodicity of elements.

[1+2]

- i. The modern periodic table is based on periodic law. State the law.
ii. Periodicity of elements is due to electronic configuration. Explain.

- b. Name the factors that affect the value of ionization energy. Why is there a steep rise in IE from carbon to nitrogen?

[2]

4. Concentrated sulphuric acid can be used in the laboratory to produce hydrogen chloride gas by the reaction with solid sodium chloride.

- i. Hydrogen iodide is not produced by the same method as hydrogen chloride. Why?

[1]

- ii. How could you identify the bottle containing HCl using ammonia gas?

[1]

- iii. How HCl is prepared? Write reaction.

[1]

- iv. Write one chemical reaction to show that conc. H_2SO_4 acts as
a. an oxidizing agent b. dehydrating agent

[2]

5. Ostwald's process is the chemical process for making nitric acid and is closely associated with the Haber's process, which provides the requisite raw material, ammonia (NH_3).

[2+1+2]

- i. How would you perform the test of the nitrate ion (NO_3^-) in its aqueous solution?
ii. What is the passivity of iron?
iii. How does ammonia react with CuSO_4 solution?

6. a. Phosphine is a colourless, flammable, very toxic gas compound with the chemical formula PH_3 , classed as a pnictogen hydride.
- How phosphine is prepared from white phosphorous? [1]
 - Write the action of phosphine on a) air b) CuSO_4 solution. [2]
- b. Differentiate between the bleaching action of Cl_2 and SO_2 . Write the reactions involved in detection of Cl^- ion in its aqueous solution? [1+1]

OR

Carbon and oxygen are two elements that are essential to life. They are naturally present in the atmosphere in the form of carbon dioxide and gaseous oxygen.

K_2O	ZnO	NO	NO_2	Pb_3O_4	Na_2O_2
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- Identify the acidic oxide, basic oxide, neutral oxide and mixed oxide from the above table. [2]
 - What happens when CO gas react with
 - Tollen's reagent
 - Cl_2 gas [2]
 - Write any one industrial applications of oxygen gas. [1]
7. a. Isomerism is very common phenomenon in organic compounds. It can be broadly classified as structural and stereoisomerism.
- Define the term Isomerism. [1+2]
 - Write the functional isomers of $\text{C}_2\text{H}_4\text{O}_2$ with IUPAC names.
- b. Write the IUPAC name of the following organic name: [2]
- $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{Br}$
 - $$\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{CH}_2-\text{CH}_2-\text{C}-\text{OH} \\ | \\ \text{Cl} \end{array}$$

8. Organic chemistry is the study of the structure, properties, composition, reaction and preparation of carbon-containing compounds.
- a. What is functional group? [1]
 - b. Write the characteristics of homologous series. [2]
 - c. Write the first four member of homologous series of aldehyde with their IUPAC names. [2]

Group - C

Give long answer to the following questions.

[3 × 8 = 24]

9. a. For the reaction,



2.5 gram of a sample of NaHCO_3 when strongly heated gives 310 cc of CO_2 gas at 27°C and 760 mmHg pressure. [2+1]

- i. Calculate the percentage purity of the sample (NaHCO_3)
 - ii. How many moles of water are produced?
- b. Draw a labeled diagram to show different spectral series of hydrogen spectra. [2]
- c. What is a Quantum number? Explain the Principal and Azimuthal quantum number. [1+2]

OR

Chemistry is a branch of science that deals with the composition, structure, and properties of substances and the change they undergo.

- a. The electroplating of silver on a spoon surface can be done by using AgNO_3 solution as electrolyte. Name the anode and cathode used during the electrolytic process? State Faraday's Second law of electrolysis. Find the mass of sodium metal deposited at cathode when 0.1 ampere of current is passed for half an hour if the process has 75% efficiency.

[1+1+2]

b. Differentiate between Gravity separation and magnetic separation methods. [2]

c. What is positive inductive effect? Arrange the following acids in the increasing order of acidic strength.

ClCH_2COOH , CH_3COOH , HCOOH [2]

10. a. After crushing and pulverization, gangue present in the ores is removed by physical and chemical methods to increase their concentration.

i. What is concentration or ore-dressing in metallurgy? [1]

ii. What is the fact on which froth floatation is based? What type of ores are concentrated by this method? Give an example of the ore. [1+0.5+0.5]

b. Nitric acid is used in the production of ammonium nitrate for fertilizers, making plastics, and in the manufacture of dyes. It is also used for making explosives such as nitroglycerin and TNT.

i. Write the principle of the manufacture of nitric acid from ammonia. [3]

ii. Write the reaction of conc. nitric acid with
a) Copper b) Hydrogen sulphide [2]

11. a. Lassaigne's test can be performed for the detection of hetero-elements present in organic compounds by fusing the compound with active metal like sodium. Write the reactions involved in the detection of Sulphur in the organic compound. [2]

- b. Define carbocation and free radical with an example of each. [2]
- c. Write the structure of following IUPAC names: [2]
- i. Propane-1,2,3-triol
 - ii. 2-chlorohexane
- d. What is valence shell and valence electron? Write the resonating structure of ozone. [2]

❧ - The End - ❧